

PROVISIONAL PATENT APPLICATION

**INDEPENDENT STRUCTURAL QUALIFICATION AND
REPRESENTATION ENTITLEMENT SYSTEMS AND
METHODS**

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Abstract

Systems and computer-implemented methods are disclosed for independently qualifying computational representations and determining what scientific, predictive, operational, regulatory, or comparative uses are supported by structural evidence. A representation-engineering subsystem generates or publishes one or more candidate representations. Each candidate is frozen or otherwise made immutable for evaluation. An independent qualification subsystem receives or replays the candidate without modifying it and computes a multidimensional qualification profile including one or more of reproducibility, robustness, stability, consistency, coverage, uncertainty, transfer, null-model separation, structural recurrence, or limitation boundaries. The subsystem produces an immutable qualification record linked to the candidate by provenance. Based on a qualification profile and a predefined qualification policy, the system determines one or more permitted interpretations or uses. In some embodiments, multiple qualified representations whose evidence is indistinguishable within a predefined tolerance form an entitlement region. An engineering policy may then select a representation from the entitlement region according to complexity, computational cost, latency, interpretability, memory, robustness, or another non-scientific preference without altering the scientific status of other entitled representations. The architecture separates candidate generation, scientific qualification, entitlement determination, and engineering selection, thereby distinguishing what evidence supports from which supported implementation is preferred. The methods may operate with recursive indexing, graph, manifold, topological, state-transition, machine-learning, scientific, biological, autonomous, or other structured computational representations.

1. Technical Field

The present disclosure relates generally to scientific computing, machine learning validation, model governance, representation analysis, reproducible computation, structural diagnostics, scientific instrumentation, decision support, and high-consequence computational systems.

More particularly, the disclosure relates to systems and methods for independently evaluating structural representations, preserving immutable evidence, determining entitlement for downstream interpretation or use, and separating scientific support from engineering preference.

2. Background

Modern computational systems frequently optimize representations and predictive models using the same data, objectives, or evaluation loops that are later used to justify deployment. This can blur the distinction between engineering a candidate solution and independently determining what the evidence supports.

Predictive performance alone does not establish whether the underlying representation is reproducible, stable, robust to perturbation, transferable across independent observations, structurally coherent, or sufficiently supported for a particular scientific or operational claim.

In many workflows, validation results are stored as transient metrics rather than as permanent scientific objects linked to a specific immutable representation. If the underlying representation continues to change after evaluation, later users may be unable to determine exactly what was evaluated or whether a reported metric still applies.

Existing model-validation and governance systems provide useful accuracy, robustness, calibration, fairness, provenance, and audit capabilities. However, a need remains for an architecture in which representation engineering and independent structural qualification are separate responsibilities, every qualification result is permanently linked to an immutable candidate, and downstream interpretation or use is constrained by explicit evidentiary entitlement.

A further problem occurs when multiple candidate representations are statistically or structurally indistinguishable within the available evidence. Selecting only the numerically highest-scoring representation may incorrectly convert measurement noise into scientific preference. The inventors recognized that scientific support can instead be represented as an entitlement region containing multiple supported candidates, after which a separate engineering policy can select among the supported candidates according to complexity, cost, latency, interpretability, or another explicitly non-scientific preference.

The architecture disclosed herein formalizes these distinctions and provides a computational evidence chain from immutable representation candidate through independent qualification, entitlement determination, and policy-governed downstream use.

3. Summary of the Disclosure

In one aspect, a representation-engineering subsystem creates or identifies a candidate computational representation and publishes or freezes the candidate so that the candidate cannot be modified during independent evaluation.

In another aspect, a qualification subsystem is logically, procedurally, or organizationally independent from the subsystem that generated the candidate. The qualification subsystem may independently replay the candidate from a snapshot, construction recipe, source data, or immutable representation product.

In another aspect, the qualification subsystem computes a multidimensional qualification profile rather than only a single score. Qualification dimensions may include reproducibility, robustness, stability, consistency, coverage, uncertainty, temporal transfer, cross-dataset transfer, null-model separation, neighborhood recurrence, transport recurrence, hierarchy recurrence, regime recurrence, or limitation boundaries.

In another aspect, each qualification run creates a qualification record that references the exact representation product evaluated, qualification profile used, metrics, thresholds, software versions, source data, random seeds, confidence measures, limitations, provenance, and resulting entitlement.

In another aspect, entitlement is purpose-specific. A representation may be entitled for exploratory observation while not entitled for publication, cross-domain comparison, autonomous action, production deployment, or regulatory evidence.

In another aspect, an entitlement region is formed from a plurality of qualified representations whose evidence is equivalent or indistinguishable within a predefined tolerance. The entitlement region preserves all supported candidates rather than elevating a single candidate solely because it has the numerically highest point estimate.

In another aspect, a separate engineering selection policy chooses among representations within the entitlement region based on an explicit preference such as minimum complexity, minimum latency, minimum memory, minimum energy, maximum interpretability, maximum robustness, minimum operational risk, or another criterion.

In another aspect, qualification cannot increase the scientific entitlement of evidence that was not evaluated, and engineering selection cannot increase scientific entitlement beyond that established by qualification.

In another aspect, qualification may itself be recursive. A structural hierarchy, topology model, generator model, comparison report, or other higher-order scientific object may undergo a new qualification process before it supports increasingly abstract interpretation.

In another aspect, the system preserves an evidence chain linking each downstream scientific or operational claim to the representation product, qualification record, entitlement decision, and policy under which the claim or use was permitted.

Core inventive combination: The disclosure is directed to the separation and coordination of four responsibilities: candidate generation, independent qualification, entitlement determination, and engineering selection. The architecture preserves immutable evidence between these stages and explicitly distinguishes what is scientifically supported from which supported implementation is preferred.

4. Brief Description of the Drawings

FIG. 1. Overall architecture separating representation engineering, immutable candidate publication, independent qualification, entitlement, engineering selection, and downstream use.

FIG. 2. Example scientific object chain linking candidate representation, qualification record, entitlement decision, and downstream use.

FIG. 3. Example independent replay workflow from immutable candidate to qualification result.

FIG. 4. Example multidimensional qualification profile.

FIG. 5. Example purpose-specific entitlement levels.

FIG. 6. Example structural entitlement region containing multiple statistically or structurally equivalent candidates.

FIG. 7. Example policy-based selection from an entitlement region.

FIG. 8. Example layered qualification of representation, hierarchy, topology, and generator objects.

FIG. 9. Example evidence chain for prediction, publication, intervention, or regulatory use.

FIG. 10. Example implementation using representation runtime, recursive structural engine, and independent scientific instrument.

FIG. 11. Example qualification-profile registry for exploratory, publication, production, regulatory, and cross-domain use.

FIG. 12. Example immutable historical record in which later qualification creates new records rather than modifying prior evidence.

5. Definitions and Interpretive Conventions

Term	Meaning
Candidate representation	A computational representation proposed for evaluation. It may be generated by any representation-learning, recursive, graph, geometric, topological, probabilistic, symbolic, or other method.
Representation engineering	The activity of generating, modifying, tuning, comparing, or proposing candidate representations.
Immutable representation product	A frozen, versioned, append-only, content-addressed, or otherwise non-modifiable representation artifact used as the object of

	qualification.
Independent qualification	Evaluation performed without modifying the candidate representation and under a responsibility, process, code path, system boundary, or governance rule separate from candidate engineering.
Qualification profile	A predefined set of required qualification categories, metrics, thresholds, confidence rules, documentation requirements, and entitlement rules for an intended purpose.
Qualification record	A persistent record describing the exact candidate evaluated, methods used, results, confidence, limitations, provenance, and entitlement outcome.
Entitlement	The set or level of interpretations, claims, predictions, actions, comparisons, or uses supported by the available qualified evidence.
Structural entitlement	Entitlement based at least in part on structural evidence such as neighborhood support, regime membership, transport stability, recursive consistency, or related organizational properties.
Scientific entitlement	Entitlement governing scientific interpretation or claims.
Operational entitlement	Entitlement governing prediction, production use, autonomous action, control, intervention, model switching, escalation, or other operational behavior.
Entitlement region	A set of representations whose evidence satisfies a qualification requirement and whose performance or structural support is mutually equivalent or indistinguishable within a predefined tolerance.
Engineering selection policy	A rule used to choose among already entitled candidates according to a preference that does not itself increase scientific support.
Evidence tolerance	A statistical, structural, numerical, confidence, equivalence, non-inferiority, reproducibility, or other predefined criterion used to determine when candidate differences are not sufficiently supported to justify scientific preference.
Evidence chain	A traceable relationship from source observation

through representation, qualification, entitlement, selection, and downstream claim or use.

Independent replay

Reconstruction or recomputation of a candidate representation or qualification result from frozen inputs, provenance, code/configuration, and recorded state.

6. Overall System Architecture



The logical separation shown above does not require separate companies, machines, or people. Independence may be implemented through separate software modules, services, execution environments, frozen inputs, access-control policies, governance rules, independent replay, or other mechanisms sufficient to prevent candidate engineering from silently altering the evidence being evaluated.

7. Scientific Object and Evidence Model

Object	Role
Representation Product	Immutable candidate structural representation.
Qualification Record	Persistent evidence describing how the representation was evaluated and with what result.

Entitlement Decision	Purpose-specific determination of supported interpretation or use.
Selection Record	Engineering policy and reason used to select among supported candidates.
Downstream Use Record	Optional record linking a claim, prediction, intervention, publication, or deployment decision to the evidence chain.

No later object need overwrite an earlier object. A later qualification result may supersede an earlier interpretation while preserving the historical qualification record and the exact representation product to which it applied.

8. Qualification Framework

Qualification category	Illustrative question
Reproducibility	Can repeated construction or replay recover comparable organization?
Robustness	Does organization remain coherent under reasonable perturbation, resampling, noise, or parameter variation?
Stability	Does the representation remain structurally coherent over time, repeated observation, or operating condition?
Consistency	Do related datasets, time segments, scales, or representations produce compatible organization?
Coverage	Does the representation characterize the observed system without excessive collapse, unsupported gaps, or unexplained regions?
Transfer	Does structural organization persist across independent temporal segments, datasets, systems, or conditions?
Null separation	Does the observed structure differ meaningfully from a randomized, permuted, shuffled, or otherwise null representation?
Structural recurrence	Do neighborhoods, bottlenecks, transport pathways, basins, hierarchy, or regimes recur across independent evaluations?
Uncertainty	What confidence or uncertainty attaches to the

metrics, structures, boundaries, or conclusions?

Limitations

Under what conditions does the representation fail to qualify or cease to support intended use?

8.1 Qualification profiles

- Exploratory research profile requiring limited evidence sufficient only for hypothesis generation.
- Scientific publication profile requiring stronger reproducibility, robustness, and documented limitations.
- Production monitoring profile requiring operational stability, update discipline, and failure boundaries.
- Regulatory or high-consequence profile requiring stricter provenance, independent replay, predefined thresholds, and use restrictions.
- Cross-domain comparison profile requiring compatible qualification evidence for all compared structures.

8.2 Qualification record schema

```
QualificationRecord {  
    record_id  
    representation_product_id  
    qualification_profile_id  
    qualification_engine_version  
    source_data_refs[]  
    replay_manifest_ref  
    metrics[]  
    confidence_intervals[]  
    robustness_results[]  
    transfer_results[]  
    null_results[]  
    limitations[]  
    entitlement_outcome  
    evaluator_or_execution_identity  
    timestamps  
    provenance_refs[]  
}
```

9. Independent Replay and Evaluation

Independent qualification preferably operates on a frozen representation product or on a deterministic replay manifest rather than on a mutable working representation.

1. Receive an immutable representation product or a replay manifest.
2. Verify object identity, integrity hash, source-data references, and configuration.
3. Reconstruct or load the candidate under an evaluation environment independent from the candidate-engineering loop.
4. Execute the predefined qualification profile.
5. Record all qualification outputs and limitations without modifying the candidate.
6. Create a qualification record linked to the exact candidate version.
7. Optionally repeat the replay in another environment, with another random initialization, or on an independent dataset.

9.1 Forms of independence

- Process independence: evaluation runs after candidate parameters are frozen.

- Code-path independence: engineering and qualification use separate modules or services.
- Data independence: qualification uses held-out temporal segments, datasets, perturbations, or resamples.
- Personnel or organizational independence: a separate reviewer or group controls qualification policy.
- Execution independence: replay occurs in a separate environment or from immutable artifacts.
- Governance independence: qualification thresholds cannot be altered by the candidate-generating process after results are observed.

10. Entitlement Determination

Qualification evidence is mapped to one or more permitted uses. Entitlement may be categorical, hierarchical, continuous, conditional, or profile specific.

Example level	Illustrative permitted use	Illustrative restriction
Exploratory	Generate structural hypotheses	No strong scientific or operational claim
Emerging	Internal comparison and follow-up studies	Additional independent evidence required
Qualified	Scientific interpretation within documented boundaries	Do not exceed limitation conditions
Reference	Reusable structural reference frame	Subject to continuing version/provenance control
Operationally Entitled	Prediction or action within validated regime	Abstain, escalate, or switch outside supported regime

These names are non-limiting. A qualification profile may define different levels or may directly map evidence to permitted operations.

11. Structural Entitlement Region

A preferred embodiment avoids converting small unsupported score differences into scientific preference.

Let R_i denote qualified candidate representations and Q_i denote qualification evidence.

Let Q_{best} denote the strongest supported evidence level.

```
EntitlementRegion E = {
   $R_i$  :
     $R_i$  satisfies the qualification profile
    AND  $difference(Q_i, Q_{best})$  is within predefined evidence tolerance
}
```

The evidence tolerance may be based on confidence intervals, equivalence margins, non-inferiority margins, bootstrap distributions, structural-similarity thresholds, paired comparisons, Bayes factors, posterior probability, permutation tests, or any other predefined criterion appropriate to the evidence.

11.1 Selection after entitlement

```
Scientific step:
  determine E = all scientifically supported candidates
```

```
Engineering step:  
  select R* = Policy(E)
```

```
Example policies:  
  minimum complexity  
  minimum latency  
  minimum memory  
  minimum energy  
  maximum interpretability  
  maximum robustness  
  lowest operational risk  
  lowest implementation cost
```

The selected representation does not become more scientifically entitled merely because an engineering policy selected it. Other candidates may remain equally entitled.

12. Entitlement-Region Algorithm

```
INPUT: qualified candidates R[1..n], qualification records Q[1..n],  
       evidence tolerance epsilon, engineering policy P
```

1. Exclude candidates that fail required qualification categories.
 2. Determine reference evidence level Q_{ref} .
 3. For each remaining candidate i :
 compute supported difference δ_i relative to Q_{ref}
 if δ_i is within epsilon:
 add R_i to entitlement region E
 4. Store E and the evidence rule used to construct E.
 5. Apply engineering policy P only to members of E.
 6. Store selection record containing P, selected R^* , and alternatives in E.
- ```
OUTPUT: entitlement region E, selected representation R^* , evidence chain
```

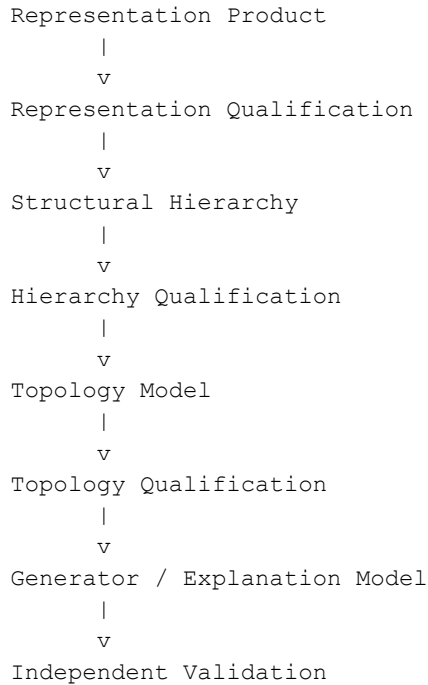
## 13. Purpose-Specific Permissions and Restrictions

- Permit exploratory visualization but not causal interpretation.
- Permit scientific comparison within one domain but not cross-domain generalization.
- Permit prediction only within a validated structural regime.
- Permit autonomous action only when both representation qualification and current operating-state entitlement are satisfied.
- Require abstention or escalation when a representation or operating state lies outside a qualified region.
- Permit use for internal engineering while withholding publication or regulatory claims.
- Permit reuse of a qualified representation product while requiring new qualification after a material representation change.

### 13.1 Entitlement cannot exceed evidence

A policy may enforce monotonicity: downstream use cannot be assigned a stronger evidentiary level than the weakest required upstream object. For example, a topology model derived from an exploratory representation cannot automatically acquire publication-level entitlement merely because a later analysis appears stable.

## 14. Layered Qualification of Higher-Order Scientific Objects



Each higher-order object may require its own qualification because increasing abstraction introduces new assumptions. Qualification of a lower-level representation does not automatically qualify a generator, theory, or intervention derived from it.

## 15. Separation of Engineering and Scientific Evaluation

One embodiment uses a representation-engineering subsystem and an independent scientific instrument. The engineering subsystem may diagnose, generate, tune, compare, or optimize candidate representations. The scientific subsystem evaluates frozen candidates under predefined qualification profiles.

```
Engineering:
 generate candidates
 improve representation
 propose alternatives
 publish immutable candidates
```

```
Science:
 replay candidates
 qualify evidence
 construct entitlement region
 record limitations
```

```
Engineering preference:
 choose among entitled candidates
```

Science does not tune the candidate.

Engineering does not rewrite qualification evidence.

## 16. Example AIIA / RIX / SII / RSD Embodiment

```
AIIA Platform / Representation Engineering
|
v
Immutable Representation Product
|
v
SII / Independent Qualification
 reproducibility | robustness | stability | consistency | coverage
|
v
Qualification Record
|
v
Structural / Scientific Entitlement
|
v
Entitlement Region
|
v
Explicit Engineering Selection Policy
|
v
RSD / next evidence-grounded investigation
```

Recursive Indexing (RIX) may be the structural computation engine used to generate or measure neighborhoods, basins, transport, topology, entropy, bottlenecks, hierarchy, or regime state. Primitive Indexing (PIX) may provide representation-runtime lifecycle and snapshot services. These named systems are non-limiting embodiments.

## 17. Operational Structural Entitlement Embodiment

A previously developed embodiment evaluates whether a current system state remains within a structurally supported operating regime before permitting prediction or autonomous action. Structural criteria may include neighborhood support, basin membership, transport drift, path accessibility, structural excursion, reentry, or coordinate-structure divergence.

```
Current Observation
|
Structural Representation
|
Compare with validated structural regime
|
Entitlement Decision
 +--> Predict / Act
 +--> Abstain
 +--> Escalate
 +--> Switch Model
 +--> Acquire Data
```

The present disclosure extends beyond this operational gating embodiment by providing the broader architecture through which representations and structural regimes are independently qualified, preserved as evidence, and assigned purpose-specific entitlement.

## 18. Comparative Qualification and Cross-Representation Evaluation

Comparisons between representations preferably inherit the qualification limitations of the compared objects.

- Compare two representations only after both satisfy a required qualification profile.
- Record differences in qualification confidence as part of the comparison.
- Prevent comparison from increasing the entitlement of either input representation.
- If one candidate is exploratory and another is qualified, label any comparison accordingly.
- Construct an entitlement region only among candidates meeting the same or explicitly compatible evidentiary requirements.

## 19. Example Structural Qualification Protocol

```
QualificationProfile: ScientificPublication
 reproducibility:
 independent seeds >= required_count
 neighborhood agreement >= threshold
 robustness:
 bootstrap stability >= threshold
 parameter sensitivity within bounds
 temporal_transfer:
 independent time block required
 null_model:
 structural separation required
 uncertainty:
 confidence interval or bootstrap distribution required
 limitations:
 mandatory
 output:
 Qualified | QualifiedWithLimitations | Exploratory | Concern
```

Thresholds and categories are illustrative and may vary by domain or use.

## 20. Evidence Chain and Claim Traceability

```
Source observations
|
Representation Product RP-042
|
Qualification Record QR-117
|
Entitlement Decision ED-23
|
Selection Record SR-08
|
Prediction / Publication / Intervention / Comparison
|
Claim or use references RP-042 + QR-117 + ED-23
```

A downstream system can therefore determine not only what representation was used, but what evidence supported its use and which limitations applied at the time.

## 21. Versioning, Supersession, and Continuing Qualification

- A change to a published representation creates a new representation product version rather than modifying the qualified object.
- A new qualification run creates a new qualification record linked to the applicable representation version.
- A later qualification result may supersede an earlier entitlement decision without deleting the historical record.
- Production systems may require periodic requalification after time, data shift, structural change, software change, or policy change.
- Cross-version comparison can identify whether a representation improved, degraded, or changed entitlement boundaries.

## 22. Example Applications

| Application                     | Illustrative use                                                                                                                  |
|---------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| Scientific discovery            | Qualify organizational atlases before scientific interpretation or recursive discovery.                                           |
| Pharmaceutical AI               | Separate representation quality evidence from downstream predictive model validation and restrict claims to qualified conditions. |
| Autonomous systems              | Permit action only when both representation qualification and current structural operating regime support action.                 |
| World models                    | Qualify latent structural organization before using structural excursions or recovery patterns operationally.                     |
| Biological representations      | Compare alternative protein representations while separating scientific support from engineering preference.                      |
| Regulatory workflows            | Preserve immutable evidence, independent replay, limitations, and use permissions.                                                |
| Enterprise AI governance        | Attach representation-level evidence to systems that otherwise report only model-level performance.                               |
| Cross-domain structural science | Require compatible qualification before comparing organizational topology across unrelated domains.                               |

## 23. Detailed Algorithmic Embodiments

### 23.1 Algorithm A — Independent qualification

INPUT: immutable representation product  $R$ , qualification profile  $P$

1. verify  $R$  identity and provenance
2. initialize independent evaluation environment
3. execute each required category in  $P$
4. record metric values, uncertainty, and failures
5. derive qualification outcome under frozen rules
6. generate immutable qualification record  $Q$
7. link  $Q$  to  $R$  without altering  $R$

OUTPUT:  $Q$

### 23.2 Algorithm B — Purpose-specific entitlement

INPUT: qualification record  $Q$ , purpose  $U$ , entitlement policy  $E$

1. identify evidence requirements for  $U$
2. compare  $Q$  against required categories and thresholds
3. if all mandatory requirements satisfied:  
    grant entitlement for  $U$  subject to recorded limitations  
    else:  
        deny, restrict, or downgrade entitlement
4. record entitlement decision and supporting  $Q$

OUTPUT: entitlement decision

### 23.3 Algorithm C — Entitlement region and engineering selection

INPUT: qualified representations  $R_i$ , records  $Q_i$ , tolerance epsilon, policy  $P$

1. determine supported candidate set  $S$
2. establish comparison reference among  $S$
3. include each candidate within tolerance epsilon in  $E$
4. record entitlement region  $E$
5. apply engineering policy  $P$  only to  $E$
6. record selected candidate and unselected entitled alternatives

OUTPUT:  $E$ , selection record

### 23.4 Algorithm D — Requalification after change

on material representation or environment change:

- create new representation product version
- identify qualification categories invalidated by the change
- rerun required qualification categories
- create new qualification record
- recalculate entitlement
- preserve all prior records

## 24. Controls, Reproducibility, and Governance

- Freeze qualification thresholds before reviewing the candidate result when confirmatory evidence is intended.
- Record every representation version and qualification-engine version.
- Require that failed qualification categories remain in the record rather than being discarded.
- Prevent candidate-engineering code from modifying immutable qualification records.
- Distinguish exploratory parameter search from independent replay.

- Use held-out datasets, time segments, bootstrap resamples, null models, or independent seeds when required by the qualification profile.
- Record limitations even when the overall qualification outcome is favorable.
- Do not permit an engineering selection policy to modify the underlying qualification evidence.
- Do not automatically generalize entitlement from one purpose, domain, or operating regime to another.

## 25. Alternative Embodiments and Variations

- Qualification may operate on a complete representation or on a snapshot, structural hierarchy, topology, generator, comparison report, or other derived scientific object.
- Independence may be logical rather than physical and may be enforced through immutable artifacts and frozen policies.
- Entitlement may be binary, ordinal, continuous, probabilistic, set-valued, or purpose-specific.
- An entitlement region may be defined using equivalence, non-inferiority, confidence-overlap, bootstrap, permutation, structural similarity, Bayesian, or other evidence rules.
- Engineering selection may use one or multiple criteria and may be deterministic, optimized, or policy based.
- Qualification records may be stored in a database, object store, distributed ledger, version-control system, scientific repository, or other persistent store.
- The system may support human approval before entitlement becomes active.
- Operational entitlement may be combined with live structural state monitoring so that a representation remains generally qualified but a particular current state is temporarily not entitled for prediction or action.
- The architecture may be implemented in cloud, local, embedded, distributed, laboratory, regulated, autonomous, or enterprise environments.

## 26. Technical Advantages

- Separates representation engineering from scientific evaluation.
- Prevents silent modification of the object being qualified.
- Preserves exact evidence through immutable representation products and qualification records.
- Supports purpose-specific interpretation rather than one generic pass/fail score.
- Prevents small unsupported score differences from becoming false scientific preference.
- Allows engineering optimization only after scientific support has been established.
- Makes limitations and failure boundaries first-class outputs.
- Supports cumulative, reproducible scientific evidence across representation versions.
- Provides traceability from downstream prediction or claim back to the representation and qualification evidence that supported it.

## 27. Illustrative Claim-Style Invention Statements

1. A computer-implemented method comprising receiving an immutable candidate representation generated by a representation-engineering subsystem; independently evaluating the candidate under a predefined qualification profile without modifying the candidate; generating a qualification record; and determining, from the qualification record, at least one permitted downstream use.



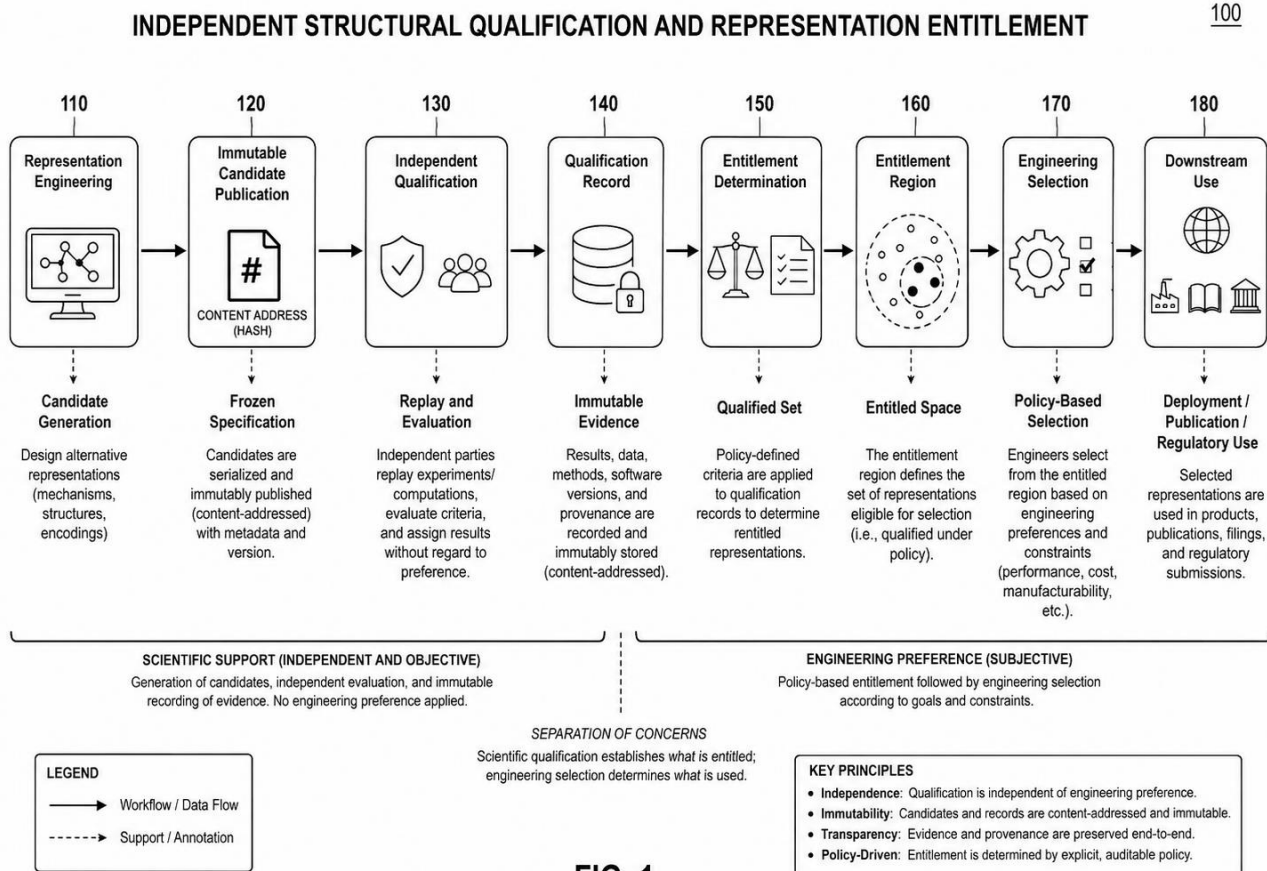
- 2.** The method above wherein the qualification profile includes at least two of reproducibility, robustness, stability, consistency, coverage, transfer, null-model separation, structural recurrence, uncertainty, or limitation analysis.
- 3.** The method above wherein the candidate representation is reconstructed by independent replay from a frozen snapshot or replay manifest.
- 4.** A method comprising storing a qualification record that identifies the exact representation product version, qualification engine version, source data, metrics, confidence measures, limitations, and entitlement outcome.
- 5.** A method comprising assigning different entitlement levels to different downstream purposes based on the same qualification record.
- 6.** A method wherein a representation is entitled for exploratory scientific use but not entitled for autonomous action or regulatory evidence.
- 7.** A computer-implemented method comprising receiving qualification records for multiple candidate representations; determining a subset of the candidate representations that are scientifically supported within a predefined evidence tolerance; and storing the subset as an entitlement region.
- 8.** The method above further comprising selecting one candidate from the entitlement region according to an engineering policy that does not modify the scientific entitlement of other members of the entitlement region.
- 9.** The method above wherein the engineering policy selects according to complexity, latency, memory, energy, interpretability, robustness, implementation cost, or operational risk.
- 10.** A method wherein a numerically highest-scoring candidate is not selected as scientifically preferable when its difference from another candidate is not supported beyond a predefined evidence tolerance.
- 11.** A system comprising a candidate-generation subsystem, an immutable representation-product store, an independent qualification subsystem, an entitlement engine, and a selection-policy engine.
- 12.** A system wherein the candidate-generation subsystem is prevented from altering qualification results produced by the independent qualification subsystem.
- 13.** A method comprising independently qualifying a higher-order structural object derived from a previously qualified representation before permitting claims based on the higher-order structural object.
- 14.** A method comprising preventing a downstream object from receiving an entitlement level exceeding a required upstream evidence level.
- 15.** A method comprising creating a new representation-product version and a new qualification record after a material representation change while preserving prior qualification records.
- 16.** A method comprising controlling prediction, abstention, escalation, model switching, or action based on a structural entitlement decision linked to a previously qualified representation.
- 17.** A method comprising requiring both representation-level qualification and current-regime entitlement before permitting an autonomous operation.
- 18.** A method comprising recording a downstream use with references to a representation product, qualification record, and entitlement decision.
- 19.** A method comprising comparing two structural representations only when both satisfy a predefined compatible qualification profile.
- 20.** A computer-readable medium storing instructions that cause one or more processors to perform any method described herein.

- 21.** A method wherein qualification thresholds are frozen before confirmatory evaluation and failed qualification categories are preserved in the qualification record.
- 22.** A method wherein a separate human authorization is required before a new entitlement level becomes active.
- 23.** A method wherein the qualification subsystem produces an entitlement decision with explicit limitations rather than only a scalar confidence score.
- 24.** A method comprising preserving multiple mutually entitled candidates and recording the non-scientific policy used to select one for implementation.

## **28. Relationship to Existing Portfolio and New Matter**

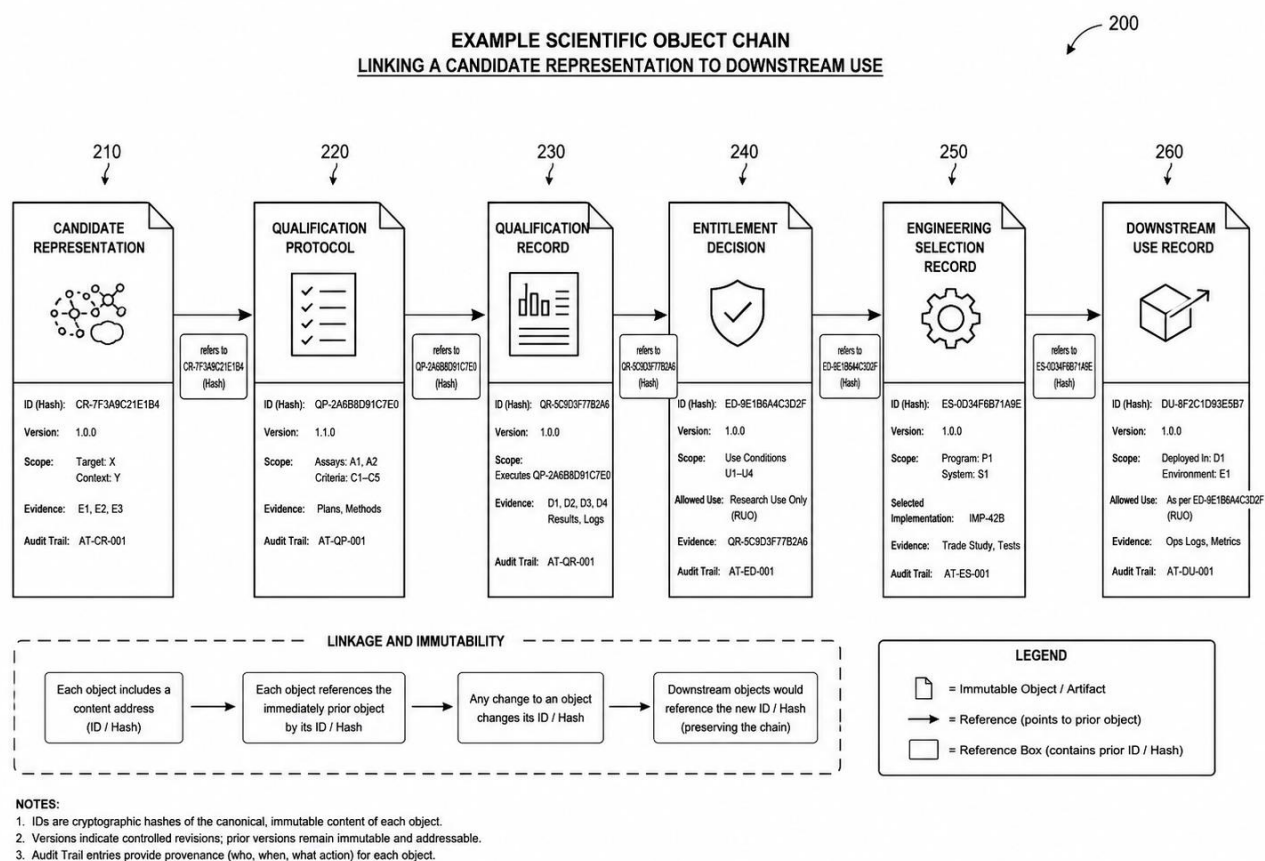
Earlier portfolio work includes recursive indexing, constraint-governed representations, computational entropy, coordinate-structure divergence, structural excursion/recovery, and structural entitlement for prediction or action. The July internal memoranda also documented representation improvement, generator entitlement, validated structural regimes, and the principle that a model should not be allowed to claim more than its demonstrated structural support.

The present disclosure focuses on broader and more explicit upper-stack architecture: independent qualification as a distinct computational responsibility; immutable qualification records; purpose-specific scientific entitlement; entitlement regions containing multiple scientifically supported candidates; and separation of engineering selection from scientific support.



**FIG. 1**

**FIG. 1.** Overall architecture separating representation engineering, immutable candidate publication, independent qualification, entitlement, engineering selection, and downstream use.



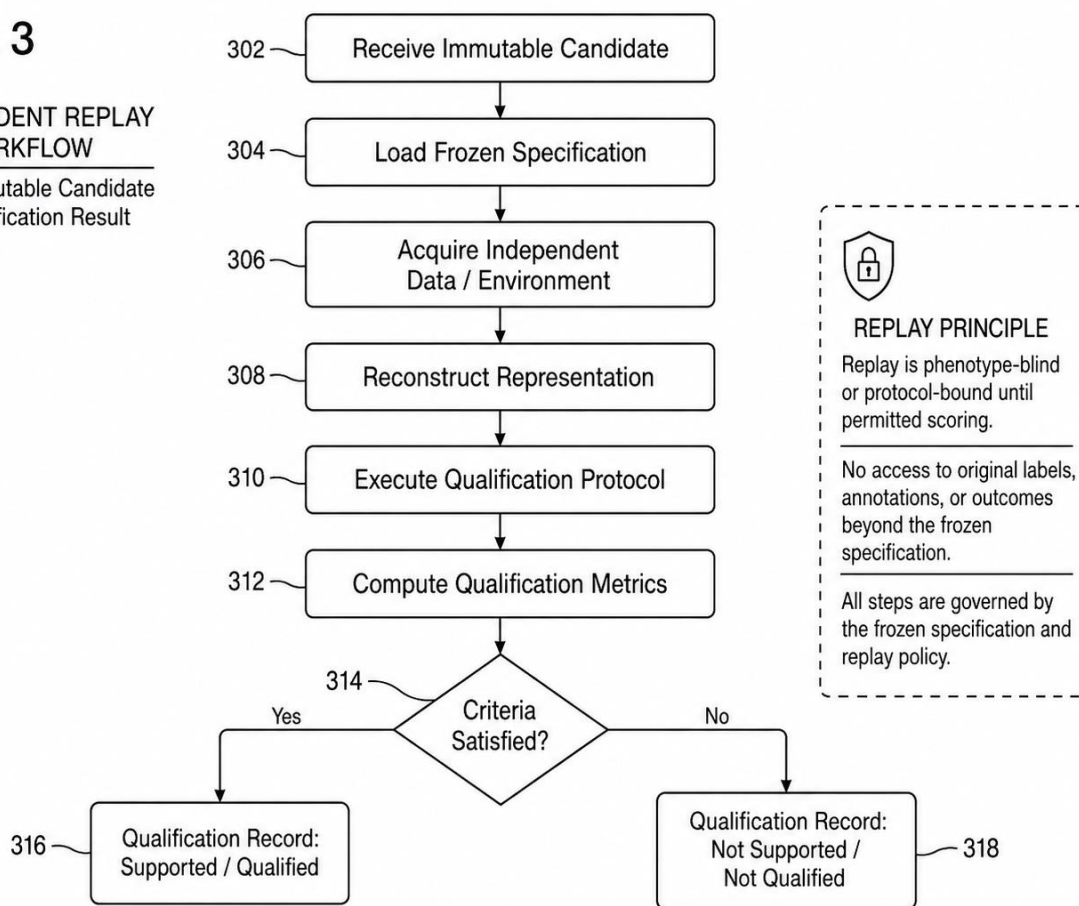
**FIG. 2**

**FIG. 2.** Example scientific object chain linking candidate representation, qualification record, entitlement decision, and downstream use.

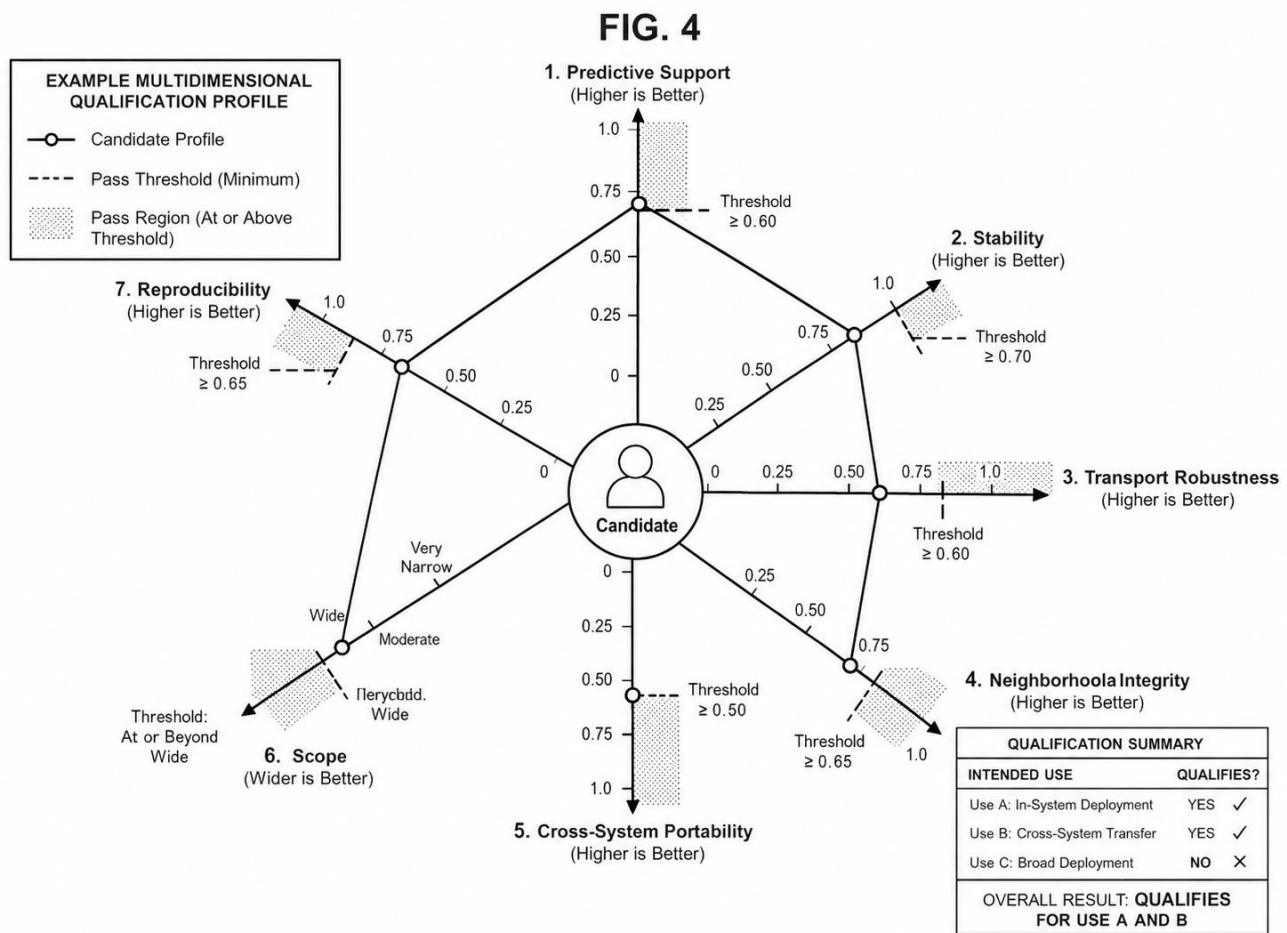
**FIG. 3**

**INDEPENDENT REPLAY  
WORKFLOW**

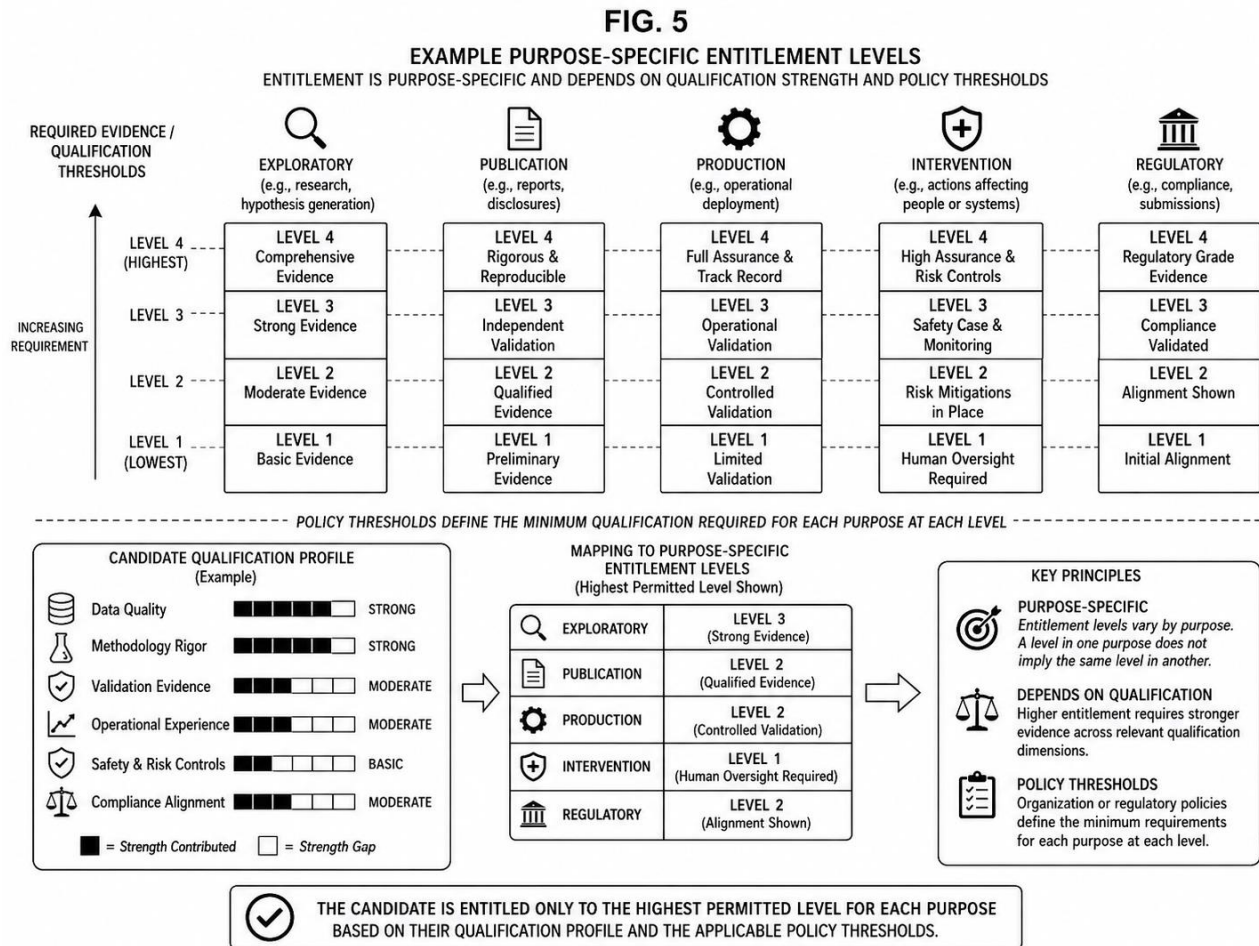
From Immutable Candidate  
to Qualification Result



**FIG. 3.** Example independent replay workflow from immutable candidate to qualification result.



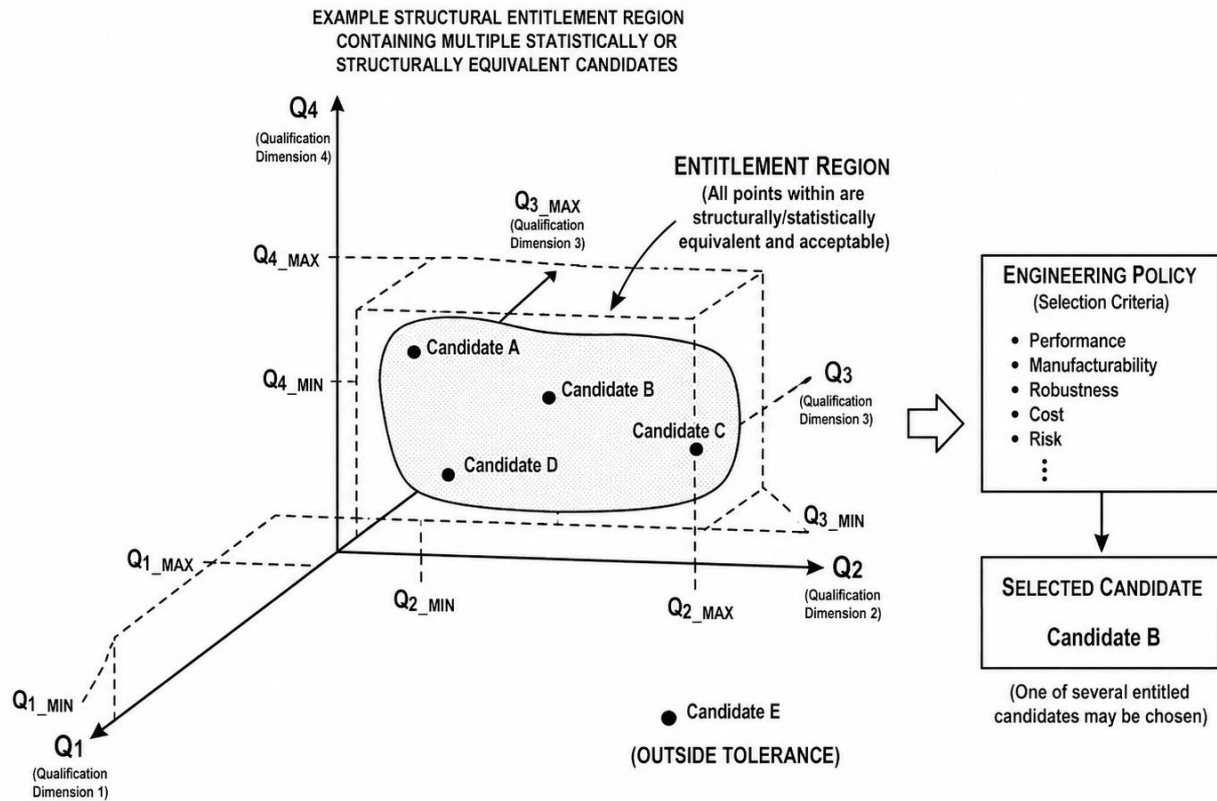
**FIG. 4.** Example multidimensional qualification profile.



**FIG. 5.** Example purpose-specific entitlement levels.



**FIG. 6**



**FIG. 6.** Example structural entitlement region containing multiple statistically or structurally equivalent candidates.



FIG. 7

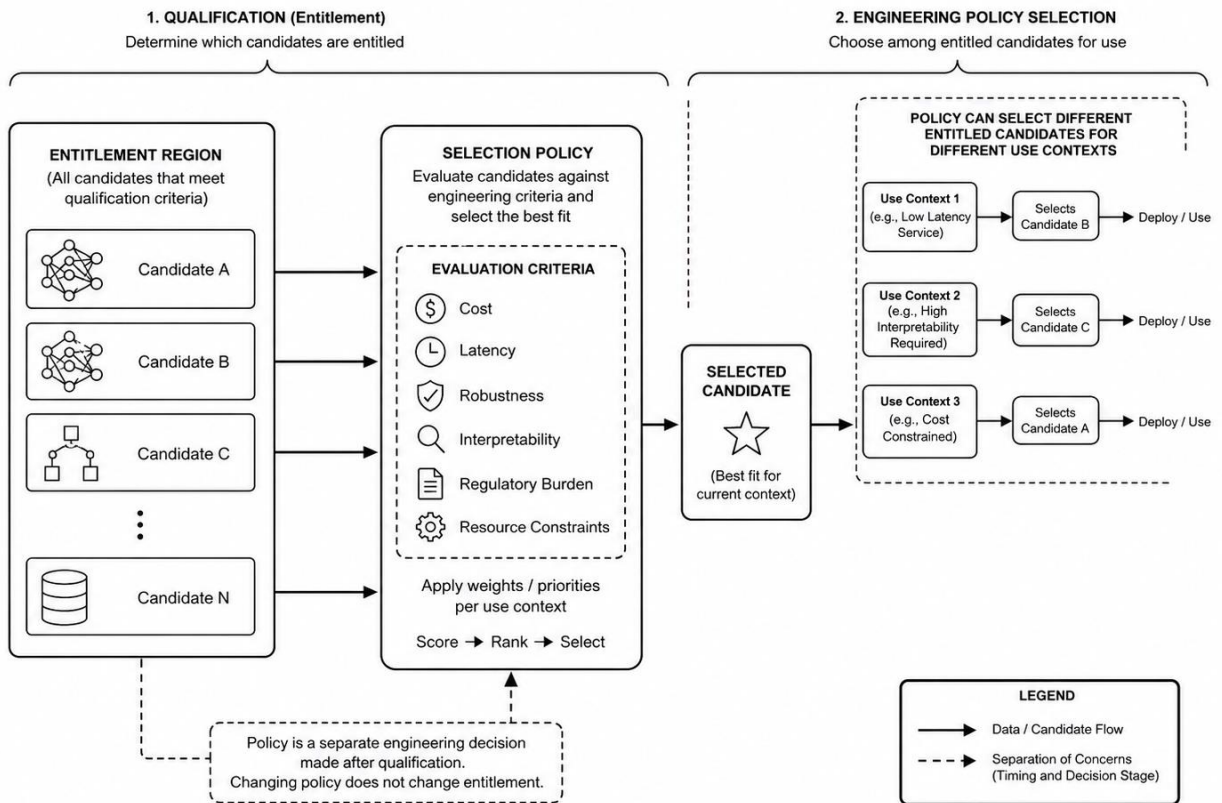
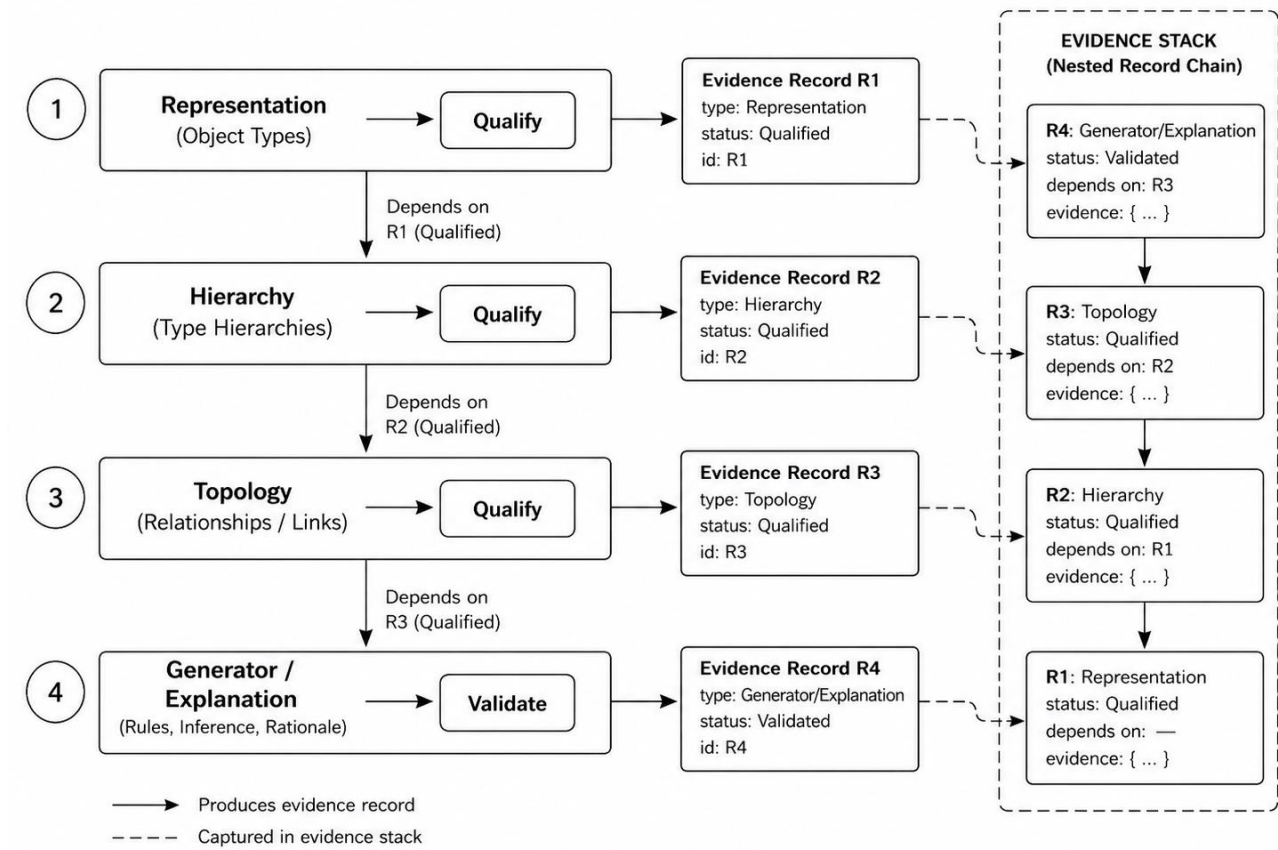
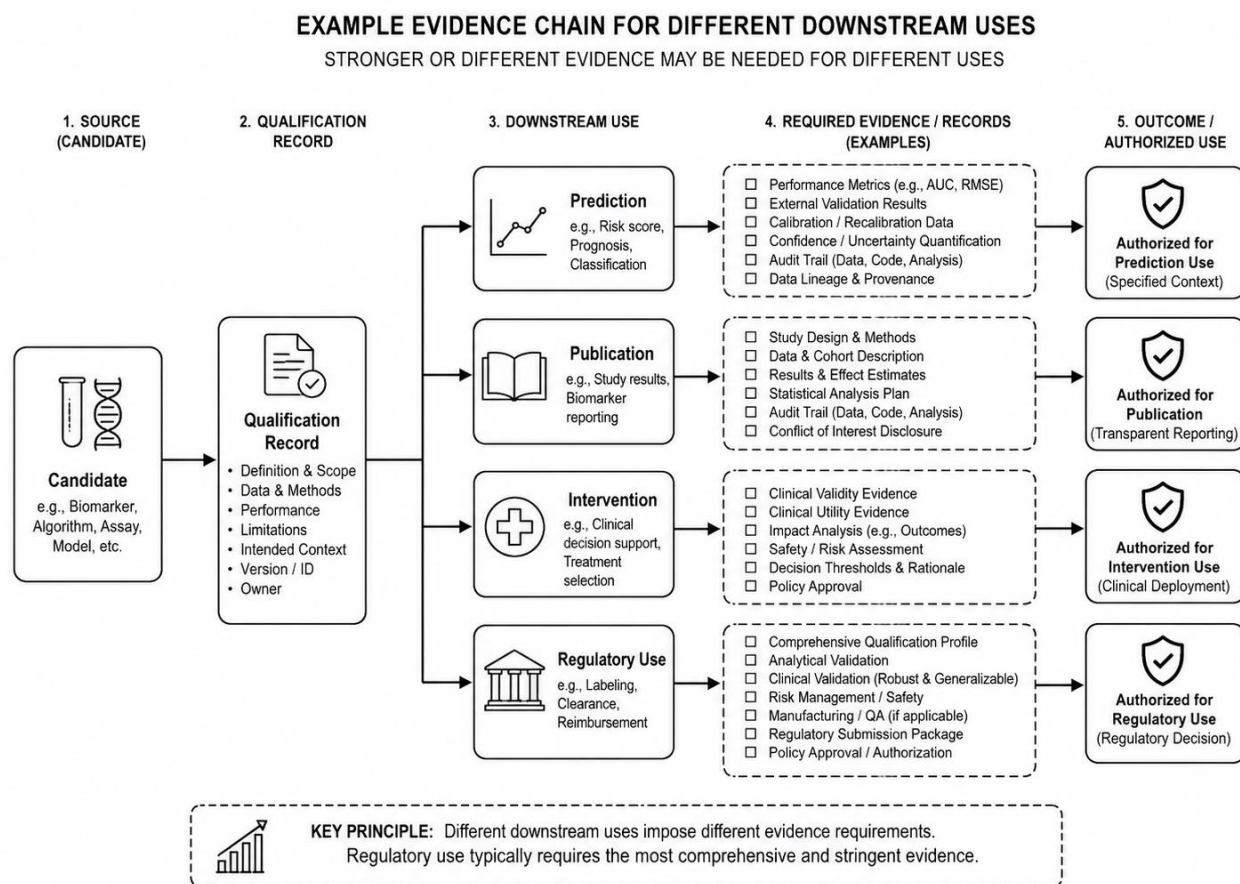


FIG. 7. Example policy-based selection from an entitlement region.



**FIG. 8**

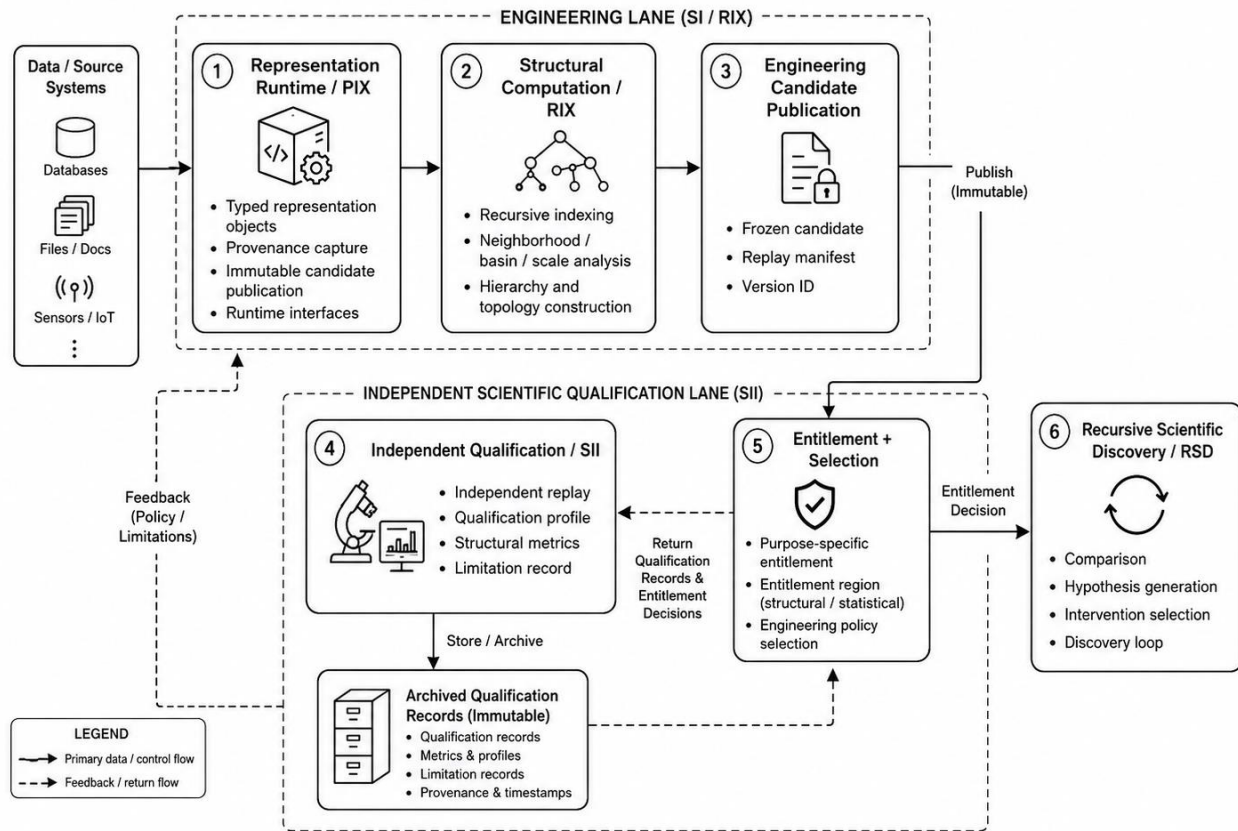
**FIG. 8.** Example layered qualification of representation, hierarchy, topology, and generator objects.



**FIG. 9**

**FIG. 9.** Example evidence chain for prediction, publication, intervention, or regulatory use.

**FIG. 10 — Example implementation using representation runtime, recursive structural engine, and independent scientific instrument**



**FIG. 10.** Example implementation using representation runtime, recursive structural engine, and independent scientific instrument.

FIG. 11 — EXAMPLE QUALIFICATION-PROFILE REGISTRY FOR EXPLORATORY, PUBLICATION, PRODUCTION, REGULATORY, AND CROSS-DOMAIN USE

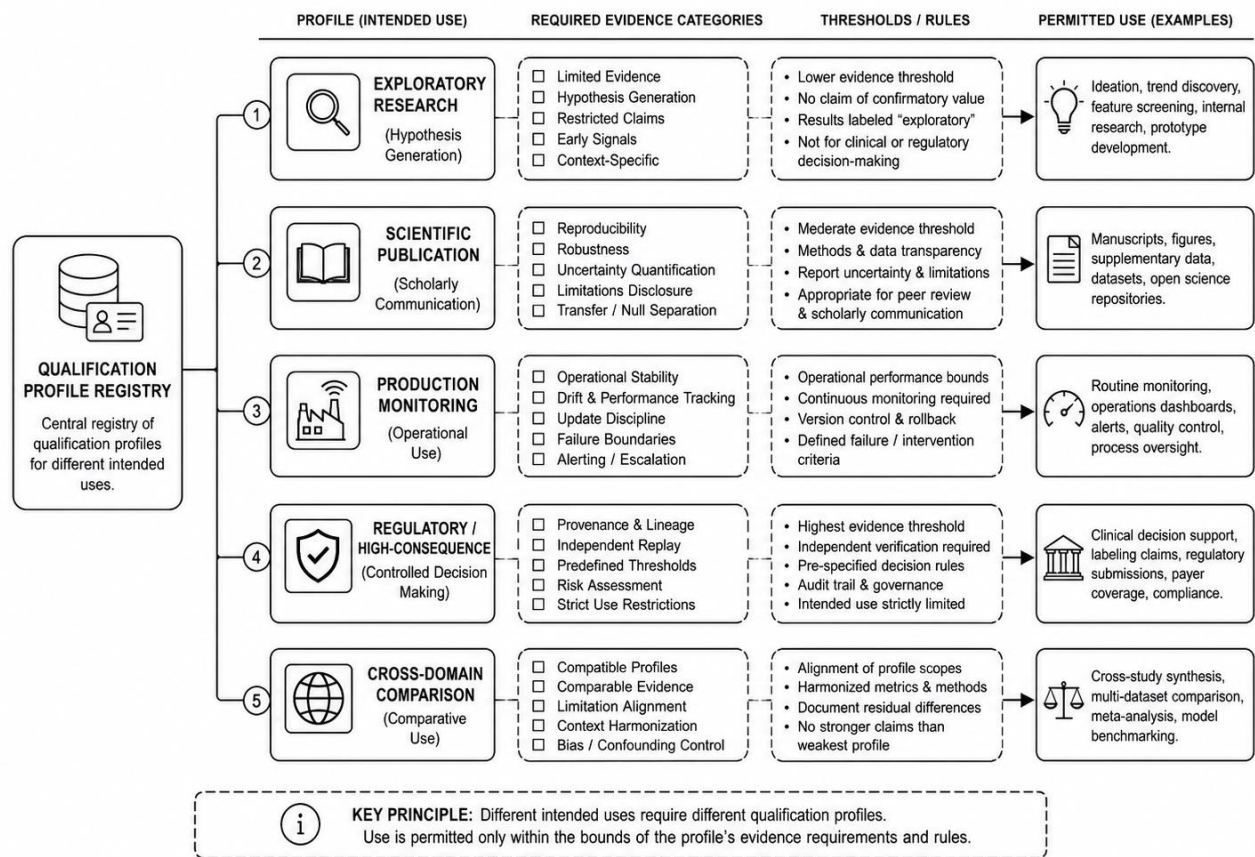
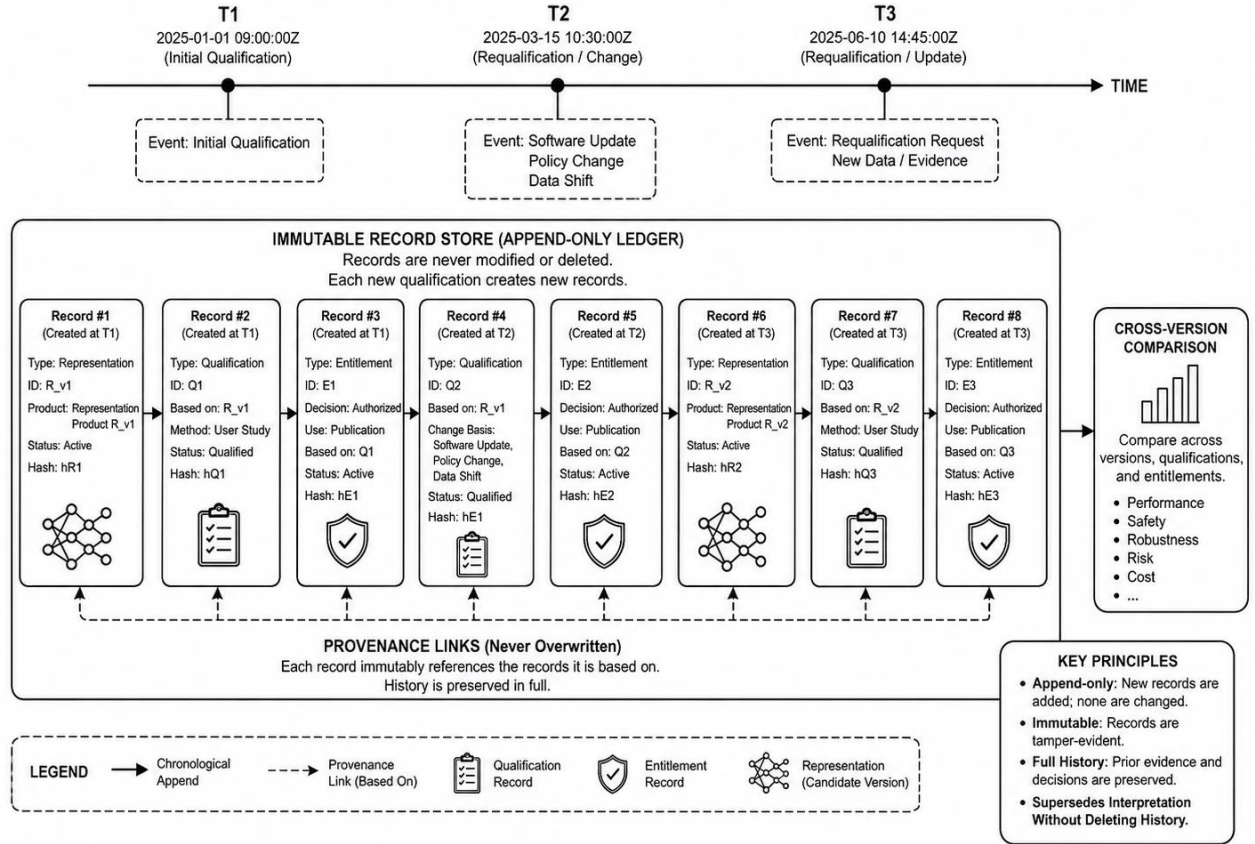


FIG. 11. Example qualification-profile registry for exploratory, publication, production, regulatory, and cross-domain use.



**FIG. 12 — Example immutable historical record in which later qualification creates new records rather than modifying prior evidence.**



**FIG. 12.** Example immutable historical record in which later qualification creates new records rather than modifying prior evidence.

## Appendix B — Example Qualification and Entitlement Policy Schemas

```
QualificationProfile {
 purpose: SCIENTIFIC_PUBLICATION
 required_categories: [reproducibility, robustness, transfer, null_separation]
 thresholds: predefined
 uncertainty_method: paired_bootstrap
 limitations_required: true
 independent_replay_required: true
}
```

```
EntitlementPolicy {
 purpose: CROSS_DOMAIN_COMPARISON
 minimum_qualification_level: QUALIFIED
 compatible_profiles_required: true
 evidence_tolerance: predefined
 selection_policy: MINIMUM_COMPLEXITY_WITHIN_ENTITLEMENT_REGION
}
```